# Monopoly

Answer key with worked solutions

Single-price monopoly: choose MR = MC, then use demand to find price. Qm is monopoly output; Qc is competitive output. For linear demand and MC, DWL = ½(Qc - Qm)[Pm - MC(Qm)].

Source: Monopoly (1).pdf

| # | Answer | Work / explanation |
|---|---|---|
| **1** | **B - $125** | TR = (200 - 3Q)Q = 200Q - 3Q², so MR = 200 - 6Q. MC = 2Q. Set MR = MC: 200 - 6Q = 2Q, giving Q = 25. Substitute into demand: P = 200 - 3(25) = $125. |
| **2** | **e - None** | A profit-maximizing single-price monopolist with downward-sloping demand chooses MR = MC, with P > MR = MC. Price does not generally equal AVC or fixed cost. |
| **3** | **B**<br>**0 < Q < 400** | Rearrange Q = 800 - 5P to get P = 160 - 0.2Q. TR = 160Q - 0.2Q², so MR = 160 - 0.4Q. Revenue increases when MR > 0: 160 - 0.4Q > 0, giving 0 < Q < 400. Revenue peaks at Q = 400. |
| **4** | **b - $400** | MR = 100 - 4Q; MC = 20. Monopoly: 100 - 4Q = 20 gives Qm = 20 and Pm = 100 - 2(20) = $60. Competition: 100 - 2Q = 20 gives Qc = 40. DWL = ½(40 - 20)(60 - 20) = $400. |
| **5** | **b - $P_{AC}$**<br>**(no subsidy)** | At $P_{AC}$, price equals average cost, so the firm covers all economic costs and earns zero economic profit. This is the lowest sustainable uniform price without a subsidy. $P_{PC}$ (choice c) gives efficient output where P = MC, but P < AC, so the firm needs a subsidy to cover its loss. The question should specify the subsidy or break-even assumption. |
| **6** | **a, b, c, d, e** | All are consistent with seeking market power. (a) Safety regulations can raise entry costs. (b) A merger removes a competitor. (c) Tariffs restrict foreign competition. (d) A patent creates a legal barrier to imitation. (e) Buying a key supplier can restrict rivals' access to inputs or raise their costs. These actions can also have other motives or benefits. |
| **7** | **d - $0** | With perfect price discrimination, the firm sells every unit whose willingness to pay is at least MC. Set 180 - Q = 30 to obtain Q = 150, the efficient quantity. All gains from trade are realized, so DWL = $0.<br>*this question is mainly about price discrimination and therefore does not apply. |
| **8** | **B - $245** | MR = 150 - 20Q; MC = 10. Monopoly: 150 - 20Q = 10 gives Qm = 7 and Pm = 150 - 10(7) = $80. Competition: 150 - 10Q = 10 gives Qc = 14. DWL = ½(14 - 7)(80 - 10) = $245. |
| **9** | **a, b, c, d, e** | All five are correct under the intended interpretation: each creates barriers to entry that protect incumbent firms. |

| # | Answer | Work / explanation |
|---|---|---|
| **10** | **E - None<br>DWL = $640** | MR = 200 - 10Q; MC = 40. Monopoly: 200 - 10Q = 40 gives Qm = 16 and Pm = 200 - 5(16) = $120. Competition: 200 - 5Q = 40 gives Qc = 32. DWL = ½(32 - 16)(120 - 40) = $640, which is not listed. |
| **11** | **a - $110** | Inverse demand is P = 200 - Q, so MR = 200 - 2Q. Set MR = MC: 200 - 2Q = 20 gives Q = 90. Price is P = 200 - 90 = $110. |
| **12** | **b - $36** | MR = 120 - 4Q; MC = 6Q. Monopoly: 120 - 4Q = 6Q gives Qm = 12, Pm = $96, and MC(Qm) = $72. Competition: 120 - 2Q = 6Q gives Qc = 15. DWL = ½(15 - 12)(96 - 72) = $36. The triangle's height is the demand-MC gap at Qm. |
| **13** | **b - $400** | Same setup as Question 4. MR = 100 - 4Q = MC = 20 gives Qm = 20 and Pm = $60. P = MC gives Qc = 40. DWL = ½(40 - 20)(60 - 20) = $400. |
| **14** | **b - $P_{AC}$<br>(no subsidy)** | Same graph as Question 5. $P_{AC}$ lets the firm cover total economic costs without a subsidy. $P_{PC}$ (choice c) produces the efficient quantity, but requires a subsidy because P < AC. The answer depends on whether a subsidy is available. |
| **15** | **a, b, c, d, e** | Same as Question 6. Regulation can raise entry barriers; a merger removes a competitor; tariffs limit foreign competition; patents restrict imitation; and control of a key supplier can disadvantage rivals. All five can serve a strategy to gain or maintain market power. |